Iterative Deepening A* (IDA*)

- IDDFS estimates the optimal solution with the bounds.

$$B = 0, 1, 2, 3, \ldots, C^*$$

- Set B to $h(s_I)$ } allows us to skip iterations.

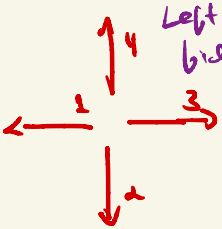
$v(n) > B$ ✓
$g(n) > B$
$h(n) + g(n) > B$

- Prune n if $f(n) > B$.

Example: IDA* with MD

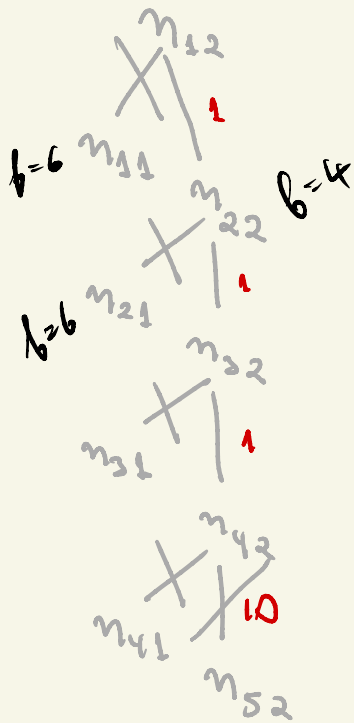
order to follow

Left action biased.



	0	1	2	3	4
0			6		
1		6	4	6	
2		6	4	6	
3		6	4	6	
4		6	4		
5			4		
4					

$B = 4$



$f(s_g) = \cancel{4} + 0 = 4$

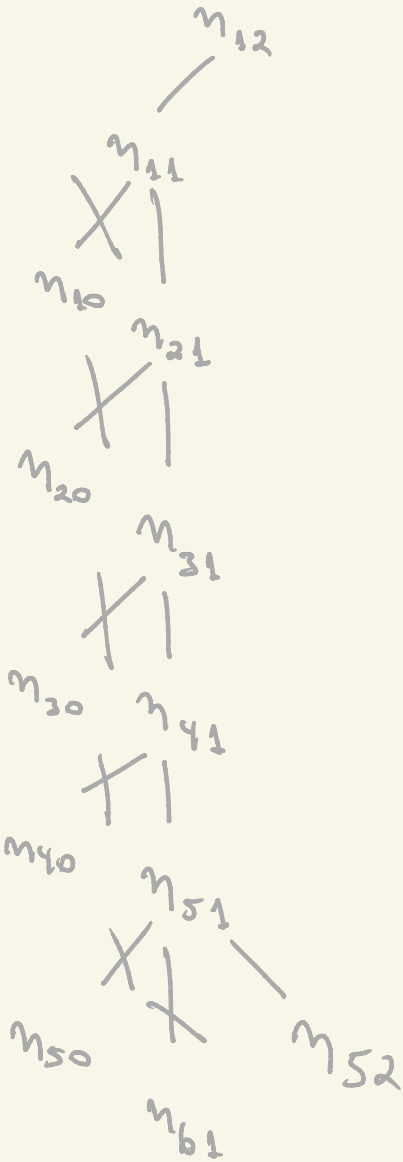
$13 + 0 = 13$

$2+6=8$

$B = \cancel{4}\ 6$

	0	1	2	3	4
0					
1	8	6	4		
2	8	6			
3	8	6			
4	8	6			
5	8	6	6		
6		8			

$6+2=8$



Only see a fraction of the tree.

IDA* Properties

- Optimal if h is admissible (sufficient but not necessary)

- Complete

- Memory Complexity: $O(d)$

- Time Complexity: ?

WA* AND WIDA*

- Multiply $h(n)$ by $w > 1$.

$$f(n) = g(n) + w\,h(n) \quad , \; w > 1$$

will find bounded suboptimal solutions.